

Software Engineering Lab 1:
EV Charging Station Reservation System:

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1. Problem Overview:

- Electric vehicle drivers require a navigation and reservation portal to locate compatible charging sockets, reserve guaranteed time slots, and calculate dynamic power rates.

2. Identify Functional Requirements:

Sl no.	Requirement	Priority	Acceptance criteria	Rationale
FR001	Match the driver's vehicle connector type with compatible chargers, and make reservations for them.	High	Given a compatible available socket, the system displays it as available and successfully creates a 30-minute reservation when selected.	Allows drivers to locate suitable charging stations conveniently.
FR002	Allow the driver to search for available charging stations based on proximity.	High	When the driver enters a valid location, the system displays at least one matching charging station when one is available within the search area	Enables the driver to select a valid charging time.
FR003	Allow the driver to view available time slots for a compatible station.	High	When a driver selects a compatible station, the system displays all currently available 30-minute slots and does not display already-reserved slots as available.	Allows drivers to know the charging cost before making a reservation
FR004	Calculate and display the charging rate and price for the same in a selected slot.	Medium	When a driver selects an available charging slot, the system calculates a rate and displays the applicable price before the reservation is confirmed.	Prevents conflicting reservations and ensures reliable slot allocation.
FR005	Confirm the reservation and prevent the reserved socket/slot from being booked concurrently by another driver.	High	When one driver successfully reserves a socket for a specific slot, a second reservation attempt for that same socket and slot is rejected.	Prevents sockets from remaining unnecessarily blocked and discourages missed reservations.

2.1. Identify Non-functional Requirements:

NFR001	The system must display available charging stations if the EV, which is connected to the app, indicates a low battery.	High	If the driver does not plug in within 10 minutes of the reservation start, the socket is released and the penalty is recorded/applied	Prevents sockets from remaining blocked, and discourages missed reservations.
NFR002	Prevent concurrent double booking of sockets by one user.	Medium	One user cannot book multiple sockets from the same car at the same time.	Prevents 3rd party users from hoarding bookings and marking them up to others
NFR003	If a charging station is down, it must not appear on the system until it is fixed.	High	Charging stations are not picked up by the app if they're offline or out of order.	Ensures a user doesn't book a faulty station.

3. Actors:

- Driver:
 - Searches for charging stations
 - Check Charger availability
 - View Charging Rates
 - Reserve a slot for charging
 - Cancel Reservation
- Station Operator:
 - Manages and maintains the charging stations
 - Views reservations
 - Updates charger availability
 - Manage Charging sockets
- Payment Gateway
 - Processes payments

4. Use Cases

ID	Use Case
UC-01	Search Charging stations
UC-02	Check Charger Availability
UC-03	View Dynamic Charging rate
UC-04	Reserve charging slot
UC-05	Process payment
UC-06	Manage Charging Station

5. Use Case Flow:

Main Success Scenario:

1. Driver enters their vehicle/connector information.
2. System identifies compatible charging stations.
3. Driver selects a charging station.
4. System displays available 30-minute charging slots.
5. Driver selects a slot.
6. System calculates the applicable dynamic charging rate.
7. System displays the reservation cost.
8. Driver confirms the reservation.
9. System locks the selected socket and time slot.
10. The system confirms the reservation with the driver.

Alternate flows:

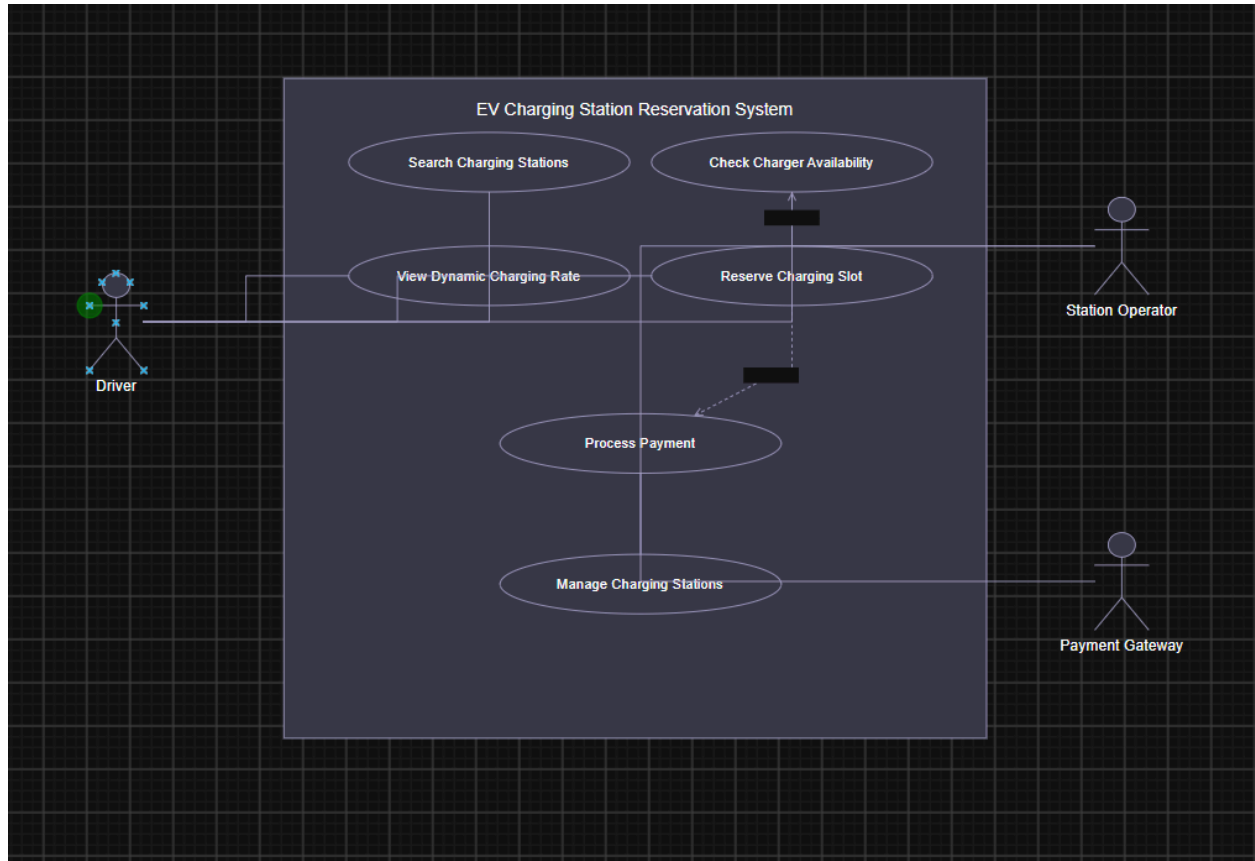
1. No Compatible Charger:

- In the event there is no compatible charging station, the system displays a message saying no compatible charger is currently available.
- The driver is prompted to search for an alternate location.
- The reservation process ends without creating a booking.

2. Slot Becomes Unavailable:

- Another driver has reserved the socket before confirmation.
- The system rejects the reservation.
- Displays other available slots.
- The driver selects another slot.

Diagram:



Use Case-Specific Preconditions & Postconditions

UC-01: Search Charging Stations

- **Preconditions:**
 - System has access to the user's current GPS location or a manually entered search location.
 - Vehicle connector type is saved in the driver's profile or selected manually.

- **Postconditions:**
 - A list of active, compatible, and proximity-matched charging stations is rendered to the user interface.

UC-02: Check Charger Availability

- **Preconditions:**
 - Driver has selected a specific charging station from the search results (UC-01).
 - Station operator status for the selected station is marked as **Online/Functional** (NFR003).
- **Postconditions:**
 - System displays all unreserved 30-minute time slots for compatible sockets at that station.
 - Currently booked or out-of-service sockets/slots are hidden or greyed out.

UC-03: View Dynamic Charging Rate

- **Preconditions:**
 - Driver has selected an available 30-minute charging slot (UC-02).
- **Postconditions:**
 - System calculates and displays the real-time dynamic tariff and total reservation cost for the selected duration prior to booking confirmation.

UC-04: Reserve Charging Slot

- **Preconditions:**
 - Selected socket and 30-minute slot are currently available in the database.
 - Driver does not have an existing active reservation for the same time window (NFR002).
- **Postconditions:**
 - Selected socket and slot are locked exclusively for the driver (FR005).
 - A 10-minute countdown timer is initiated relative to the reservation start time for auto-release enforcement (NFR001).

UC-05: Process Payment

- **Preconditions:**
 - Driver has confirmed the booking details and selected a valid payment method.
 - Payment Gateway API connection is active.

- **Postconditions:**

- Transaction is processed; a digital receipt and payment confirmation are generated.
- If payment fails, the temporary lock on the socket/slot is released immediately.

UC-06: Manage Charging Station

- **Preconditions:**

- Station Operator is authenticated with administrative credentials.

- **Postconditions:**

- Station operational state (Active/Maintenance) or individual socket configurations are updated in real time across the app.